

plan: adjoint operators (part 2), projections (part 2),
Gram matrix, non-orthonormal bases

last time: $A: X \rightarrow Y$ bounded linear operator $\left(\begin{array}{l} X, Y \text{ inner} \\ \text{product spaces} \end{array} \right)$

$A^*: Y \rightarrow X$ is defined by requiring

$$\langle Ax, y \rangle_Y = \langle x, A^*y \rangle_X \quad \text{for all } x \in X, y \in Y$$

Theorem: Every bounded linear operator has an adjoint.
(assuming X is a Hilbert space)

proof: given $y \in Y$, $l(x) = \langle Ax, y \rangle$ belongs to X' .

Riesz $\Rightarrow \exists! v$ s.t. $l(x) = \langle x, v \rangle$ for all x .

$$\text{So } \langle Ax, y \rangle = \langle x, A^*y \rangle \quad \forall x \quad (\Rightarrow) \quad \underbrace{A^*y = v.}$$

this works (the property holds)
and it determines A^*y uniquely.

linearity: given $y, z \in Y$ and $\alpha, \beta \in \mathbb{C}$

$v = A^*(\alpha y + \beta z)$ is the unique vector for which

$$\langle Ax, \alpha y + \beta z \rangle = \langle x, v \rangle \quad \forall x \in X$$

$$\begin{aligned} \text{LHS} &= \bar{\alpha} \langle Ax, y \rangle + \bar{\beta} \langle Ax, z \rangle = \bar{\alpha} \langle x, A^*y \rangle + \bar{\beta} \langle x, A^*z \rangle \\ &= \langle x, \alpha A^*y + \beta A^*z \rangle \quad \text{for all } x, \text{ so } v = \alpha A^*y + \beta A^*z \end{aligned}$$

properties: ① $(A^*)^* = A$

$$② (AB)^* = B^* A^*$$

$$③ (A^{-1})^* = (A^*)^{-1}$$

$$④ \|A^*\| = \|A\|$$

$$⑤ \text{ran}(A)^\perp = \ker(A^*)$$

proof: ① $\langle A^*y, x \rangle_X = \langle y, Ax \rangle_Y \quad \forall x \in X, y \in Y$

this is the condition for either A or A^* to be the adjoint of the other.

②, ③ see HW2

④ first we show that $\|A^*y\| \leq \|A\| \cdot \|y\| \quad \forall y \in Y$.

This implies that $\|A^*\| = \sup_{y \neq 0} \frac{\|A^*y\|}{\|y\|} \leq \|A\|$ (since $\|A\|$ is an upper bound on each ratio $\frac{\|A^*y\|}{\|y\|}$ that occurs, and "sup" is the least upper bound.)

$$\begin{aligned} \text{proof: } \|A^*y\|^2 &= \langle A^*y, A^*y \rangle = \langle y, AA^*y \rangle \\ &= |\langle y, AA^*y \rangle| \stackrel{\text{C.S.}}{\leq} \|y\| \cdot \|AA^*y\| \\ &\leq \|y\| \cdot \|A\| \cdot \|A^*y\| \end{aligned}$$

If $A^*y \neq 0$, $\|A^*y\| > 0$ and dividing by it preserves the inequality.

$$\|A^*y\| \leq \|y\| \cdot \|A\|. \leftarrow \text{This is also true if } A^*y = 0.$$

$$\text{so for all } y \in Y, \quad \|A^*y\| \leq \|A\| \cdot \|y\|. \quad \therefore \|A^*\| \leq \|A\|.$$

Reversing the roles of A, A^* we have

$$\|Ax\|^2 = \langle Ax, Ax \rangle = \langle x, A^*Ax \rangle = |\langle x, A^*Ax \rangle|$$

$$\stackrel{\text{C.S.}}{\leq} \|x\| \cdot \|A^*Ax\| \leq \|x\| \cdot \|A^*\| \cdot \|Ax\|$$

If $Ax \neq 0$, divide by $\|Ax\|$:

$$\|Ax\| \leq \|A^*\| \cdot \|x\| \quad \leftarrow \text{also true if } Ax = 0.$$

$$\therefore \|A\| \leq \|A^*\| \leq \|A\|.$$

$$\therefore \|A^*\| = \|A\|.$$

$$\textcircled{5} \quad \text{ran}(A)^\perp = \ker(A^*)$$

$$y \in \text{ran}(A)^\perp \Leftrightarrow \langle Ax, y \rangle = 0 \quad \forall x \in X$$

$$\Leftrightarrow \langle x, A^*y \rangle = 0 \quad \forall x \in X$$

$$\Leftrightarrow A^*y = 0$$

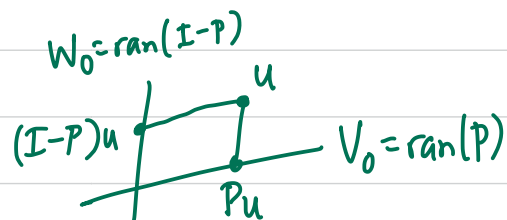
$$\Leftrightarrow y \in \ker(A^*)$$

Projections, part 2 Let V be a Hilbert space.

def: $P: V \rightarrow V$ is a projection if P is bounded and $P^2 = P$.

If P is a projection, so is $I - P$

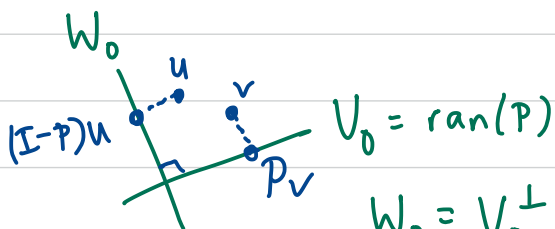
$$\text{pf: } (I - P)^2 = I - P - \underbrace{P + P^2}_{=0} = I - P$$



def: An orthogonal projection is a projection P that also satisfies $P^* = P$.

claim: $(I-P)u \perp P_v \quad \forall u, v \in V$

$$\begin{aligned} \text{pf: } \langle (I-P)u, P_v \rangle &= \langle P(I-P)u, v \rangle = \langle (P-P^2)u, v \rangle \\ &\quad \uparrow P=P^* \qquad \qquad \qquad \uparrow P^2=P \\ &= \langle 0, v \rangle = 0 \end{aligned}$$



$$W_0 = V_0^\perp = \ker(P^*) = \ker(P) = \text{ran}(I-P)$$

$\begin{aligned} Pu = 0 &\Rightarrow (I-P)u = u - 0 = u \Rightarrow u \in \text{ran}(I-P) \\ u \in \text{ran}(I-P) &\Rightarrow \exists y \in V \text{ s.t. } u = (I-P)y \\ &\Rightarrow Pu = P(I-P)y = (P-P^2)y = 0 \\ &\Rightarrow Pu = 0 \end{aligned}$
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As before, $(I-P)u \in V_0^\perp \Leftrightarrow P_u = \text{closest point to } u \text{ in } V_0 = \text{range}(P)$

$$V = V_0^\perp \oplus V_0$$

$$u = (I-P)u + Pu$$

↗ basis-independent description of P_u

recall that if $e_1, \dots, e_N$ are an O.N. basis for V_0 then $P_u = \sum_{n=1}^N \langle u, e_n \rangle e_n$

It's often useful to compute projections from a non-orthonormal basis.

Let $V_0 \subseteq V$ be a finite dimensional subspace with basis $\varphi_1, \dots, \varphi_N$.

Define

$$T: \mathbb{C}^N \rightarrow V \quad \text{by} \quad Tc = \sum_{k=1}^N c_k \varphi_k$$

The adjoint of T is the operator $T^*: V \rightarrow \mathbb{C}^N$ s.t.

$$\langle T^*f, c \rangle_{\mathbb{C}^N} = \langle f, Tc \rangle_V \quad \forall f \in V, c \in \mathbb{C}^N$$

Expand it out:

$$\sum_{j=1}^N (T^*f)_j \bar{c}_j = \langle f, \sum_j c_j \varphi_j \rangle = \sum_j \bar{c}_j \langle f, \varphi_j \rangle$$

$$\text{so } (T^*f)_j = \langle f, \varphi_j \rangle \quad \leftarrow \text{components of } T^*f \in \mathbb{C}^N$$

$$\left[\text{Alternatively: } (T^*f)_j = \langle T^*f, e_j \rangle = \langle f, Te_j \rangle = \langle f, \varphi_j \rangle \right]$$

The k th column of a matrix is the result of applying the matrix to the k th unit vector e_k in $\mathbb{C}^N$

$$\downarrow$$
$$(T^*T)_{jk} = (T^*(Te_k))_j = \langle Te_k, \varphi_j \rangle = \langle \varphi_k, \varphi_j \rangle$$

T^*T is an invertible $N \times N$ matrix since $\varphi_1, \dots, \varphi_N$ are linearly independent. (It's called the Gram matrix)

$$\text{pf: } \left(\text{If } T^*Tc = 0, \text{ then } 0 = \langle c, T^*Tc \rangle = \langle Tc, Tc \rangle \Rightarrow Tc = \sum_k c_k \varphi_k = 0 \Rightarrow c = 0 \right)$$

Claim: $P = T(T^*T)^{-1}T^*$ is the orthogonal projection onto V_0 .

$$\begin{pmatrix} | & | & | & | \\ \varphi_1 & \varphi_2 & \dots & \varphi_N \end{pmatrix} \begin{pmatrix} \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}^{-1} \begin{pmatrix} \langle \cdot, \varphi_1 \rangle \\ \vdots \\ \langle \cdot, \varphi_N \rangle \end{pmatrix}$$

$$V_0 = \text{ran}(T) = \text{span}\{\varphi_1, \dots, \varphi_N\}$$

proof: ① $P^2 = T \underbrace{(T^*T)^{-1}T^*T}_{I} (T^*T)^{-1}T^* = P$

② $P^* = T^* \underbrace{(T^*T)^{-1}}_{(ABC)^* = C^*B^*A^*} T^* = T \underbrace{(T^*T)^*}_{(A^{-1})^* = (A^*)^{-1}}^{-1} T^* = P$
 $(T^*T)^* = T^*T$

③ $\text{range}(P) = V_0$.

$$\text{ran}(P) = \text{ran}(T(T^*T)^{-1}T^*) \subseteq \text{range}(T) = V_0$$

$\uparrow$
 T is applied last

$$P\varphi_k = T \underbrace{(T^*T)^{-1}T^*}_{I} (T e_k) = T e_k = \varphi_k \leftarrow \begin{matrix} \text{each } \varphi_k \text{ is "hit" by } P\varphi_k \\ \text{(so } \leq \text{ becomes } =) \end{matrix}$$

every $f = \sum c_k \varphi_k$ is in $\text{ran}(P)$ since $Pf = f$.

Note: If $\varphi_1, \dots, \varphi_N$ is an orthonormal basis for V_0 , then

$$(T^*T)_{jk} = \langle \varphi_k, \varphi_j \rangle = \delta_{jk}$$

$$(T^*T)^{-1} = I$$

$$P = T T^* = (\varphi_1, \dots, \varphi_N) \begin{pmatrix} \langle \cdot, \varphi_1 \rangle \\ \vdots \\ \langle \cdot, \varphi_N \rangle \end{pmatrix} = \sum_{k=1}^N \langle \cdot, \varphi_k \rangle \varphi_k$$

same formula as before